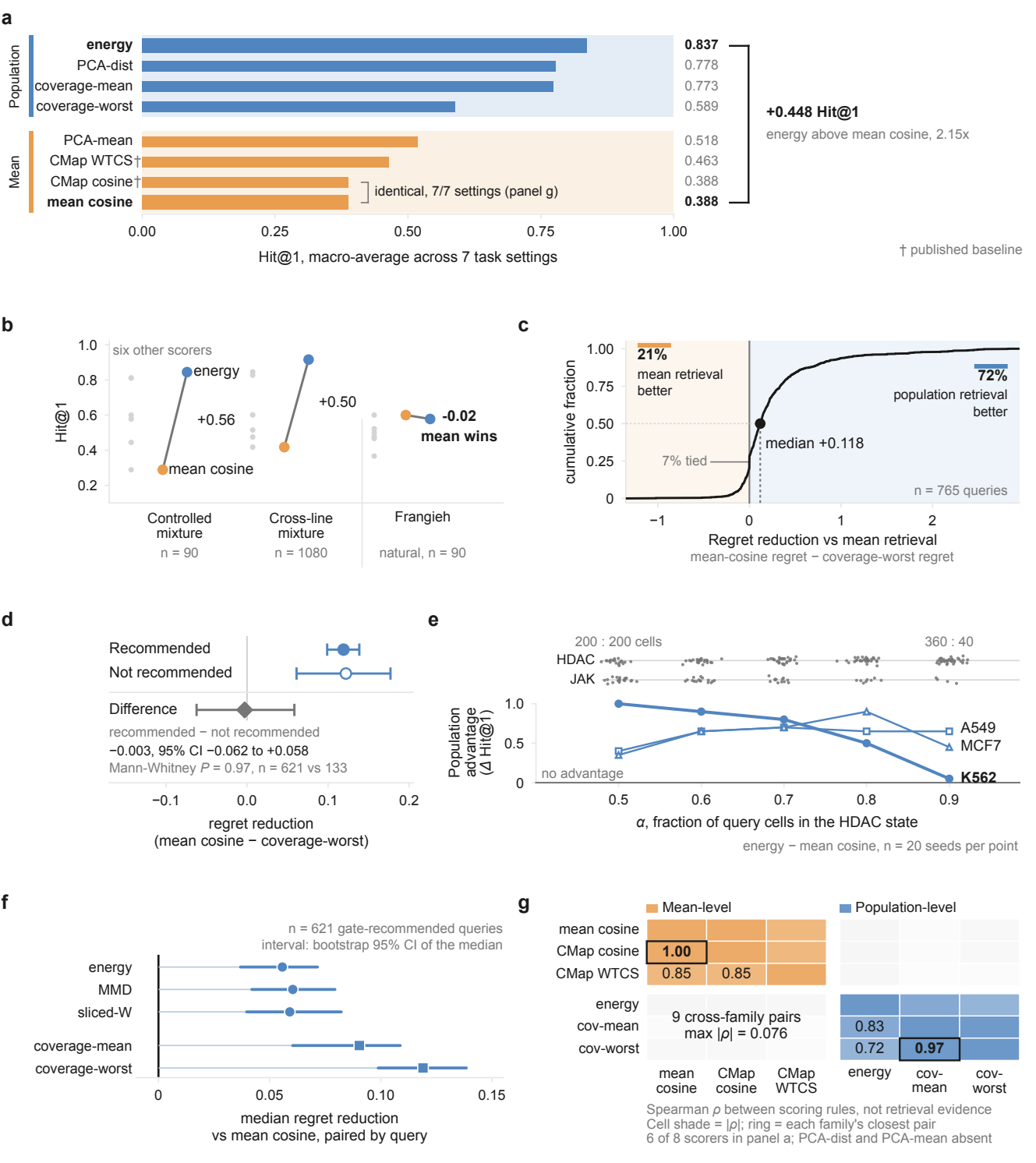

a
Population
energy
PCA-dist
coverage-mean
coverage-worst
0.837
0.778
0.773
0.589
Mean
PCA-mean
CMap WTCS†
CMap cosine†
mean cosine
0.518
0.463
0.388
0.388
identical, 7/7 settings (panel g)
+0.448 Hit@1
energy above mean cosine, 2.15x
0.00
0.25
0.50
0.75
1.00
Hit@1, macro-average across 7 task settings
† published baseline
b
Hit@1
six other scorers
energy
+0.56
mean cosine
+0.50
-0.02
mean wins
Controlled mixture
n = 90
Cross-line mixture
n = 1080
Frangieh
natural, n = 90
c
cumulative fraction
21%
mean retrieval better
72%
population retrieval better
median +0.118
7% tied
n = 765 queries
Regret reduction vs mean retrieval
mean-cosine regret − coverage-worst regret
d
Recommended
Not recommended
Difference
recommended − not recommended
−0.003, 95% CI −0.062 to +0.058
Mann-Whitney P = 0.97, n = 621 vs 133
regret reduction
(mean cosine − coverage-worst)
e
200 : 200 cells
360 : 40
HDAC
JAK
Population advantage (Δ Hit@1)
no advantage
A549
MCF7
K562
α, fraction of query cells in the HDAC state
energy − mean cosine, n = 20 seeds per point
f
n = 621 gate-recommended queries
interval: bootstrap 95% CI of the median
energy
MMD
sliced-W
coverage-mean
coverage-worst
median regret reduction
vs mean cosine, paired by query
g
Mean-level
Population-level
mean cosine
CMap cosine
CMap WTCS
energy
cov-mean
cov-worst
1.00
0.85
0.85
9 cross-family pairs
max |ρ| = 0.076
0.83
0.72
0.97
Spearman ρ between scoring rules, not retrieval evidence
Cell shade = |ρ|; ring = each family's closest pair
6 of 8 scorers in panel a; PCA-dist and PCA-mean absent